

Quiz — 2.1.2 Thinking Ahead

Name: _____ Date: _ **Mark:** ____ / 20

OCR H446 · Component 02 · 2.1.2

Section A — Multiple choice (6 marks)

1. A program performs a binary search on a list of customer IDs. The precondition that must hold before the search runs is that the list is: A. Empty B. Sorted into order C. Stored in a database D. Displayed on screen Your answer: ☐
 2. In an online shop's "place order" routine, which of the following is an **input**? A. The confirmation email sent to the customer B. The customer's chosen quantity and delivery address C. Validating the card number D. The updated stock total saved to the database Your answer: ☐
 3. A weather website recomputes the same regional forecast for thousands of visitors each minute. Storing the computed forecast for quick reuse is an example of: A. A precondition B. Decomposition C. Caching D. A race condition Your answer: ☐
 4. Which is the main trade-off a developer must accept when adding caching to the weather site? A. Cached data may become stale and serve out-of-date forecasts B. The site can never use the cache again C. Caching always makes the site slower D. Caching removes the need for inputs Your answer: ☐
 5. A studio reuses a pre-written, pre-tested physics module in three different games. A key benefit of using such a reusable component is: A. It guarantees the games run concurrently B. Faster development with fewer bugs because the module is already tested C. It removes the need for any inputs or outputs D. It makes the games' data automatically sorted Your answer: ☐
 6. Which of these is an **output** of a cash machine's "withdraw cash" routine, rather than an input or a process? A. The PIN typed by the user B. The amount requested by the user C. The dispensed banknotes and updated balance D. Checking the balance is sufficient Your answer: ☐
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Section B — Short answer (9 marks)

7. A login routine must run before a user can view their account page. State the **precondition** for displaying the account page, and explain what could go wrong if the routine ran without checking it. [2]

8. A flight-booking website caches its currency-conversion rates. Discuss **one** benefit and **one** trade-off of caching the rates in this scenario. [2]

A set of handwriting practice lines. It consists of four horizontal lines: a solid top line, a dashed midline, a solid baseline, and a dashed descender line. These lines are repeated across the width of the page to help guide letter formation and size.

This image shows a blank sheet of primary-ruled paper. It features two solid black horizontal lines near the top and bottom edges. Between these solid lines are three dashed blue horizontal lines, evenly spaced. The background is white, and there are no other markings or text on the page.

This image shows a full page of primary-ruled notebook paper. It features multiple sets of horizontal lines designed to guide young learners' handwriting. Each set consists of three lines: two solid outer lines defining the height of capital letters, and a dashed middle line indicating where lowercase letters should sit. These ruled sections are repeated down the entire page, providing ample space for practicing letter formation and basic writing skills.

